Back2U Lost & Found - Comprehensive Microservice Testing Pipeline (SE3112)

One GitHub Actions workflow runs four testing paradigms against the SE3022 microservices (Auth, Item, Matching, Admin Verify): each in its own runner, with a fresh environment where needed, and one shared reporting contract. Numbers ① to ⑥ match the explanations on page 2.

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Unit testing & mocking · CI workflow

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Mutation testing

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Load & performance testing

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UI / E2E testing

1 Triggers

- **Push** — code or pipeline files changed
- **Pull request** — into develop or main
- **Manual run** — pick a category · dev / strict

GitHub Actions

`ase-testing-pipeline.yml`

ubuntu-24.04 runners

one live run per branch
(a newer push cancels the older run)

Development mode allows unfinished suites; strict mode requires them all

2 Job 1 · Unit testing

runs first · own runner

`unit-testing/run.sh`

xUnit 2.5 + Moq 4.20 · .NET 8

dotnet test on 4 projects, one per service

523 cases in 32 stories
Auth 176 · Item 115 · Matching 136 · AV 96

Parameterised [Theory] rows and Moq doubles isolate the service logic from DB, Kafka, HTTP

No running environment needed

Output: baseline
"passed" unlocks mutation testing

TRX results → summary.md, details.json
Demo mode: paced story-by-story output
Artifact: ase-unit-testing

needs

3 Job 2 · Matrix: 3 runners in parallel

after unit, even if it failed · fail-fast off · one result per category

Mutation — Stryker.NET 4.0

`mutation-testing/run.sh`
Mutates each service's code; its unit tests must kill the mutant
2,710 mutants in scope · 891 of 1,303 tested killed
Gate per service: Auth 81% ≥ 75 · Item 66% ≥ 60 · Matching 64% ≥ 60 · Admin Verify 74% ≥ 70 · needs baseline ②

Load — JMeter 5.6.3

`load-testing/run.sh`
80 s: 20 steady users, a 20-user spike, 1 admin; 34 endpoints
Real journeys: report, edit, claim, confirm, appeal, review
Gates: errors ≤ 1%, p95 per endpoint, coverage (52 checks)
Uses the shared environment ④ · HTML dashboard

UI / E2E — Selenium 4.25

`ui-e2e-testing/run.sh`
Headless Chrome drives the real React frontend
71 cases · 14 page objects · Smoke 8 / Demo 13 / Full
Cross-service journeys; screenshot on every failure
Uses the shared environment ④ · xUnit 2.9, TRX

started for Load and UI/E2E

6 Job 3 · Summary

always runs, after both jobs

Downloads every ase-* artifact and reads each category's status.txt

Category	Status
unit-testing	passed
mutation-testing	passed
load-testing	passed
ui-e2e-testing	passed

Published as one table on the run's summary page

Run fails on any failed, blocked or missing result; strict mode also fails unfinished categories

Every result is a download

every job writes

4 Shared test environment

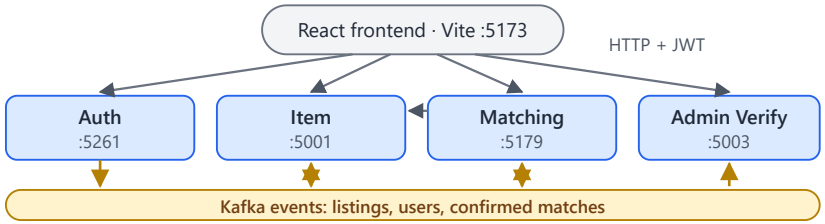
shared/start-environment.sh · fresh on every Load and UI/E2E runner · stopped after the run (logs kept)

Docker Compose (throwaway)
MySQL 8.0.40 · 4 databases
Kafka 7.5.0 (KRaft)
13 topics created before services

seed-test-data.sh <category>
optional data per category

Email, Gemini and blob storage off: nothing leaves the runner

System under test (SE3022 Back2U, Development mode, migrations + seeded accounts)



Each service has its own MySQL database · Matching also calls Item over HTTP

5 Unified reporting contract

same files from every category

Written by each run.sh

`summary.md`	scope, counts, failures
`details.json`	one case per check
`status.txt`	passed / failed / ...
`console.log`	full output

+ native: TRX, Stryker, JMeter, screenshots

shared validate-report
rejects empty or malformed reports; every case needs a name, description and status; a failed case can't exit 0
then writes status.txt

Artifact ase-<category>, kept 14 days
reports + each service's log (Load, UI/E2E) · uploaded even when a step fails

Arrows: job order / request · environment started for that job · Kafka events · reports written · reports collected

How the pipeline works

Every push or pull request that touches the services, the frontend or the pipeline starts one workflow. It checks the code at four levels, from the inside out: **unit tests** check each service's logic in isolation, **mutation testing** checks that those unit tests really catch faults, **load testing** checks the running system under many concurrent users, and **UI/E2E testing** checks complete user journeys through the real frontend. Each level is owned by one member, runs on its own machine, and reports in the same format, so one table shows the result of the whole run.

1 Triggers and workflow

- **Push** (any branch) or **pull request** into develop or main, when files under `services/`, `frontend/`, `testing-pipeline/` or the workflow change.
- **Manual run**: pick one category or all, and a mode. **Development** lets unfinished suites report "not implemented"; **strict** requires every selected suite to run and pass. Real failures fail both.
- Runs on fresh `ubuntu-24.04` GitHub runners. Only one run per branch is live: a newer push cancels the older one.

2 Job 1: unit testing and mocking

- **xUnit 2.5 + Moq 4.20** on .NET 8, one test project per service: **523 cases in 32 stories** (Auth 176, Item 115, Matching 136, Admin Verify 96).
- Parameterised `[Theory]` cases cover boundaries and input rules; Moq test doubles replace repositories, Kafka publishers and HTTP clients, so only the service logic is tested.
- Runs first and needs no environment. Its result is the **baseline**: mutation testing runs only after a passing baseline. A demo mode paces the output story by story.

3 Job 2: three stages in parallel

After unit testing finishes (pass or fail), a matrix starts three runners at once, with fail-fast off, so one stage failing never hides another's result.

Mutation (Stryker.NET 4.0). Makes small changes ("mutants") to each service's code and runs that service's unit tests against each one; a test must fail to "kill" it. One run per service, each gated on its own score, set just under its measured baseline: Auth 81% (gate 75), Item 66% (60), Matching 64% (60), Admin Verify 74% (70). 2,710 mutants in scope; untested code is reported as a coverage gap, not hidden. No environment needed.

Load (Apache JMeter 5.6.3). 80 s against the running system: 20 steady users with think time, a 20-user spike at 60 s with none, and one admin throughout. Users report, edit, photograph, resolve and delete listings, claim and confirm matches with partner accounts and send appeals, while the admin reviews spam and decides appeals: 34 endpoints over all four services. Judged on error rate (≤ 1%), p95 per endpoint and coverage (every journey ran): 52 checks.

UI/E2E (Selenium WebDriver 4.25 + xUnit 2.9). Headless Chrome drives the real React frontend, and through it all four services: 71 cases built on 14 page objects, grouped Smoke (8), Demo (13) and Full. Each test arranges its own data through the API and saves a screenshot when it fails.

4 Shared test environment

- `shared/start-environment.sh` starts throwaway **MySQL 8.0.40** and **Kafka 7.5.0** in Docker Compose, creates the four databases and the 13 Kafka topics, then builds and starts Auth, Item, Matching and Admin Verify in Development mode (migrations run, test accounts seeded) and the React frontend.
- Started fresh on every Load and UI/E2E runner, never shared or reused, and stopped afterwards with the logs kept. A category can add its own data through `seed-test-data.sh`.
- Email, Gemini image processing and blob storage are switched off, so no test reaches a real service or person.

5 Unified reporting contract

- Each category has exactly one entry point, its `run.sh`, which CI and local runs both call. It writes `summary.md` (scope, counts, failures, measurements) and `details.json` (one case per check, with a description); the runner adds `status.txt` and `console.log`.
- A shared `validate-report` rejects missing or malformed reports, and a failed case reported with exit code 0.
- Native reports travel alongside: TRX files, Stryker HTML, the JMeter dashboard, UI screenshots. Each category uploads an `ase-<category>` artifact (with service logs for Load and UI/E2E), kept 14 days.

6 Job 3: summary and final result

Always runs last. It collects every artifact, publishes one table (category and status) on the run's summary page, and sets the final result: any failed, blocked or missing category fails the run, and in strict mode so does an unfinished one.

Running it locally

The same scripts run on a developer machine (WSL2 or Linux with Docker, .NET 8, Node 22; Java 17 for load):

```
bash testing-pipeline/shared/start-environment.sh start
bash testing-pipeline/<category>/run.sh
bash testing-pipeline/shared/start-environment.sh stop
```

Unit and mutation testing need no environment. Each category folder has a README with its own commands, demo options and results.

Tools and versions

Area	Tools
CI	GitHub Actions, ubuntu-24.04, Bash, Node 22.14 (reporting)
Unit	.NET 8, xUnit 2.5, Moq 4.20
Mutation	Stryker.NET 4.0 (dotnet tool, pinned)
Load	Apache JMeter 5.6.3 (checksum-verified, cached), Java 17
UI/E2E	Selenium WebDriver 4.25, xUnit 2.9, Chrome
Environment	Docker Compose, MySQL 8.0.40, Kafka 7.5.0

Design principles

- **Isolation:** every stage on its own runner and, where needed, its own fresh environment, so results are repeatable and independent.
- **One entry point per stage:** the same `run.sh` locally and in CI, so a local pass means the same checks.
- **Fail loudly:** thresholds, mutation gates, coverage checks and report validation turn quality problems into a failed run, not a footnote.
- **No side effects:** nothing touches the deployed system, real users or real email.

Source: testing-pipeline/ (workflow .github/workflows/ase-testing-pipeline.yml). Figures from the latest validated runs, October 2026.